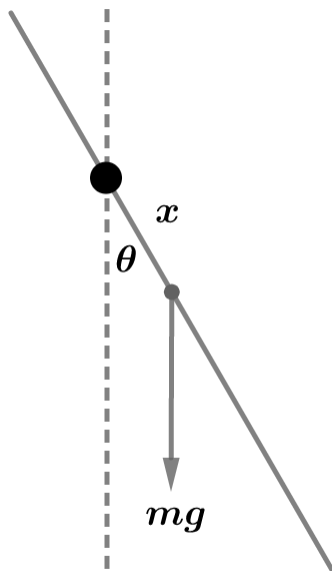


2018B F=ma Exam: Problem 21

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This is a physical pendulum so we will apply $\tau = I\alpha$:

$$-mgx \sin \theta = I\alpha$$

$$\alpha = -\frac{mgx}{I} \sin \theta$$

For small θ , the motion is approximately simple harmonic:

$$\alpha = -\frac{mgx}{I} \theta$$

$$\omega = \sqrt{\frac{mgx}{I}}$$

Hence,

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{mgx}}$$

Recall the moment of inertia of a rod about its center is

$$I_{rod} = \frac{1}{12}ML^2$$

By the parallel-axis theorem,

$$I = I_{rod} + Mx^2 = \frac{1}{12}ML^2 + Mx^2$$

To minimize the period, we can minimize I/mx :

$$\frac{d}{dx} \left(\frac{I}{mx} \right) = \frac{d}{dx} \left(\frac{L^2}{12x} + x \right) = -\frac{L^2}{12x^2} + 1 = 0$$

Thus,

$$x = \frac{L}{2\sqrt{3}}$$

so the answer is B.